伪凝聚复形零次上同调函子的仿射表示

Counting scope. The primary problem is Tag 08JX, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题在所有基变换的负上同调消失假设下证明零次上同调函子仿射有限呈示可表示，并在更强消失下得到 perfect 性。证明先做 fppf 下降，再将伪凝聚复形截成非负 Tor 振幅的有限局部自由复形，最后把函子写成矩阵线性方程的解概形。结构截断是引用输入，但函子下降与显式表示衔接仍是实质证明义务，判 H2。.

History: excerpt from revision local cached snapshot; exact chapter hashes; revision unknown, quot.tex, lines 432–581. Only reference presentation was adapted. Licensed under GNU FDL 1.2 or later; no invariant sections or cover texts. See accompanying COPYING.

Lemma 0.1. *Let R be a ring. Let $\mathfrak{p} \subset R$ be a prime ideal. Let $i \in \mathbf{Z}$. Let $K^\bullet$ be a pseudo-coherent complex of R-modules such that $H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p})) = 0$. Then there exists an $f \in R$, $f \notin \mathfrak{p}$ and a canonical direct sum decomposition*

$$K^\bullet \otimes_R R_f = \tau_{\geq i+1}(K^\bullet \otimes_R R_f) \oplus \tau_{\leq i-1}(K^\bullet \otimes_R R_f)$$

in $D(R_f)$ with $\tau_{\geq i+1}(K^\bullet \otimes_R R_f)$ a perfect complex with tor-amplitude in $[i+1, \infty]$.

证明. This is an often used special case of Lemma lemma better cut complex in two (Tag 0A1U). A direct proof is as follows. We may assume that $K^\bullet$ is a bounded above complex of finite free R-modules. Let us inspect what is happening in degree i:

$$\ldots \to K^{i-2} \to R^{\oplus l} \to R^{\oplus m} \to R^{\oplus n} \to K^{i+2} \to \ldots$$

Let A be the $m \times l$ matrix corresponding to $K^{i-1} \to K^i$ and let B be the $n \times m$ matrix corresponding to $K^i \to K^{i+1}$. The assumption is that $A \bmod \mathfrak{p}$ has rank r and that $B \bmod \mathfrak{p}$ has rank $m - r$. In other words, there is some $r \times r$ minor a of A which is not in $\mathfrak{p}$ and there is some $(m-r) \times (m-r)$-minor b of B which is not in $\mathfrak{p}$. Set $f = ab$. Then after inverting f we can find direct sum decompositions $K^{i-1} = R^{\oplus l-r} \oplus R^{\oplus r}$, $K^i = R^{\oplus r} \oplus R^{\oplus m-r}$, $K^{i+1} = R^{\oplus m-r} \oplus R^{\oplus n-m+r}$ such that the module map $K^{i-1} \to K^i$ kills of $R^{\oplus l-r}$ and induces an isomorphism of $R^{\oplus r}$ onto the corresponding summand of K^i and such that the module map $K^i \to K^{i+1}$ kills of $R^{\oplus r}$ and induces an isomorphism of $R^{\oplus m-r}$ onto the corresponding summand of K^{i+1}. Thus $K^\bullet$ becomes quasi-isomorphic to

$$\ldots \to K^{i-2} \to R^{\oplus l-r} \to 0 \to R^{\oplus n-m+r} \to K^{i+2} \to \ldots$$

and everything is clear. □

Lemma 0.2. *Let R be a ring. Let $K^\bullet$ be a bounded above complex of flat R-modules with tor-amplitude in $[a, b]$. Then $\mathrm{Coker}(d_K^{a-1})$ is a flat R-module.*

Definition 0.3. Let R be a ring. Denote $D(R)$ its derived category. Let $m \in \mathbf{Z}$.

(1) An object $K^\bullet$ of $D(R)$ is *m-pseudo-coherent* if there exists a bounded complex $E^\bullet$ of finite free R-modules and a morphism $\alpha : E^\bullet \to K^\bullet$ such that $H^i(\alpha)$ is an isomorphism for $i > m$ and $H^m(\alpha)$ is surjective.
(2) An object $K^\bullet$ of $D(R)$ is *pseudo-coherent* if it is quasi-isomorphic to a bounded above complex of finite free R-modules.
(3) An R-module M is called *m-pseudo-coherent* if $M[0]$ is an m-pseudo-coherent object of $D(R)$.
(4) An R-module M is called *pseudo-coherent*[1] if $M[0]$ is a pseudo-coherent object of $D(R)$.

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Lemma 0.4. *Let S be a scheme. Let B be an algebraic space over S. Let K be a pseudo-coherent object of $D(\mathcal{O}_B)$.*

(1) *If for all $g : T \to B$ in (Sch/B) the cohomology sheaf $H^{-1}(Lg^*K)$ is zero, then the functor*

$$(Sch/B)^{opp} \longrightarrow Sets, \quad (g : T \to B) \longmapsto H^0(T, H^0(Lg^*K))$$

is an algebraic space affine and of finite presentation over B.
(2) *If for all $g : T \to B$ in (Sch/B) the cohomology sheaves $H^i(Lg^*K)$ are zero for $i < 0$, then K is perfect, K locally has tor amplitude in $[0, b]$, and the functor*

$$(Sch/B)^{opp} \longrightarrow Sets, \quad (g : T \to B) \longmapsto H^0(T, Lg^*K)$$

is an algebraic space affine and of finite presentation over B.

证明. Under the assumptions of (2) we have $H^0(T, Lg^*K) = H^0(T, H^0(Lg^*K))$. Let us prove that the rule $T \mapsto H^0(T, H^0(Lg^*K))$ satisfies the sheaf property for the fppf topology. To do this assume we have an fppf covering $\{h_i : T_i \to T\}$ of a scheme $g : T \to B$ over B. Set $g_i = g \circ h_i$. Note that since h_i is flat, we have $Lh_i^* = h_i^*$ and h_i^* commutes with taking cohomology. Hence

$$H^0(T_i, H^0(Lg_i^*K)) = H^0(T_i, H^0(h_i^*Lg^*K)) = H^0(T, h_i^*H^0(Lg^*K))$$

Similarly for the pullback to $T_i \times_T T_j$. Since Lg^*K is a pseudo-coherent complex on T (Cohomology on Sites, Lemma lemma pseudo coherent pullback (Tag 08H4)) the cohomology sheaf $\mathcal{F} = H^0(Lg^*K)$ is quasi-coherent (Derived Categories of Spaces, Lemma lemma pseudo coherent (Tag 08JL)). Hence by Descent on Spaces, Proposition proposition fpqc descent quasi coherent (Tag 04W8) we see that

$$H^0(T, \mathcal{F}) = \mathrm{Ker}(\prod H^0(T_i, h_i^*\mathcal{F}) \to \prod H^0(T_i \times_T T_j, (T_i \times_T T_j \to T)^*\mathcal{F}))$$

In this way we see that the rules in (1) and (2) satisfy the sheaf property for fppf coverings. This means we may apply Bootstrap, Lemma lemma locally algebraic space finite type (Tag 04U0) to see it suffices to prove the representability étale locally on B. Moreover, we may check whether the end result is affine and of finite presentation étale locally on B, see Morphisms of Spaces, Lemmas lemma affine local (Tag 03WG) and lemma finite presentation local (Tag 0410). Hence we may assume that B is an affine scheme.

Assume $B = \mathrm{Spec}(A)$ is an affine scheme. By the results of Derived Categories of Spaces, Lemmas lemma pseudo coherent (Tag 08JL), lemma derived quasi coherent small etale site (Tag 071Q), and lemma descend pseudo coherent (Tag 08HE) we deduce that in the rest of the proof we may think of

[1] This clashes with what is meant by a pseudo-coherent module in Bibliography: Bourbaki-CA.

K as a perfect object of the derived category of complexes of modules on B in the Zariski topology. By Derived Categories of Schemes, Lemmas lemma pseudo coherent (Tag 08E5), lemma affine compare bounded (Tag 06Z0), and lemma pseudo coherent affine (Tag 08E7) we can find a pseudo-coherent complex $M^\bullet$ of A-modules such that K is the corresponding object of $D(\mathcal{O}_B)$. Our assumption on pullbacks implies that $M^\bullet \otimes^{\mathbf{L}}_A \kappa(\mathfrak{p})$ has vanishing H^{-1} for all primes $\mathfrak{p} \subset A$. By More on Algebra, Lemma 0.1 we can write

$$M^\bullet = \tau_{\geq 0} M^\bullet \oplus \tau_{\leq -1} M^\bullet$$

with $\tau_{\geq 0} M^\bullet$ perfect with Tor amplitude in $[0, b]$ for some $b \geq 0$ (here we also have used More on Algebra, Lemmas lemma glue perfect (Tag 066Y) and lemma glue tor amplitude (Tag 066N)). Note that in case (2) we also see that $\tau_{\leq -1} M^\bullet = 0$ in $D(A)$ whence $M^\bullet$ and K are perfect with tor amplitude in $[0, b]$. For any B-scheme $g : T \to B$ we have

$$H^0(T, H^0(Lg^* K)) = H^0(T, H^0(Lg^* \tau_{\geq 0} K))$$

(by the dual of Derived Categories, Lemma lemma negative vanishing (Tag 05TC)) hence we may replace K by $\tau_{\geq 0} K$ and correspondingly $M^\bullet$ by $\tau_{\geq 0} M^\bullet$. In other words, we may assume $M^\bullet$ has tor amplitude in $[0, b]$.

Assume $M^\bullet$ has tor amplitude in $[0, b]$. We may assume $M^\bullet$ is a bounded above complex of finite free A-modules (by our definition of pseudo-coherent complexes, see More on Algebra, Definition 0.3 and the discussion following the definition). By More on Algebra, Lemma 0.2 we see that $M = \mathrm{Coker}(M^{-1} \to M^0)$ is flat. By Algebra, Lemma lemma finite projective (Tag 00NX) we see that M is finite locally free. Hence $M^\bullet$ is quasi-isomorphic to

$$M \to M^1 \to M^2 \to \ldots \to M^d \to 0 \ldots$$

Note that this is a K-flat complex (Cohomology, Lemma lemma bounded flat K flat (Tag 06YD)), hence derived pullback of K via a morphism $T \to B$ is computed by the complex

$$g^* \widetilde{M} \to g^* \widetilde{M^1} \to \ldots$$

Thus it suffices to show that the functor

$$(g : T \to B) \longmapsto \mathrm{Ker}(\Gamma(T, g^* \widetilde{M}) \to \Gamma(T, g^* (\widetilde{M^1}))$$

is representable by an affine scheme of finite presentation over B.

We may still replace B by the members of an affine open covering in order to prove this last statement. Hence we may assume that M is finite free (recall that M^1 is finite free to begin with). Write $M = A^{\oplus n}$ and $M^1 = A^{\oplus m}$. Let the map $M \to M^1$ be given by the $m \times n$ matrix (a_{ij}) with coefficients in A. Then $\widetilde{M} = \mathcal{O}_B^{\oplus n}$ and $\widetilde{M^1} = \mathcal{O}_B^{\oplus m}$. Thus the functor above is equal to the functor

$$(g : T \to B) \longmapsto \{(f_1, \ldots, f_n) \in \Gamma(T, \mathcal{O}_T) \mid \sum g^\sharp(a_{ij}) f_i = 0,\ j = 1, \ldots, m\}$$

Clearly this is representable by the affine scheme

$$\mathrm{Spec}\left(A[x_1, \ldots, x_n]/(\sum a_{ij} x_i; j = 1, \ldots, m)\right)$$

and the lemma has been proved. $\square$